Test 1: Next Wed, Sept 24

→ Practice Test in Canvas

→ Closed book/closed notes

→ Covers transport equation & Laplace's/Poisson's equation

$G(x,y)$ in $B(0,1)$:

$$G(x,y) = \Phi(x-y) - \Phi(|x|(y-\tilde{x}))$$

$\tilde{x} = \dfrac{x}{|x|^2}$ not defined when $x=0$

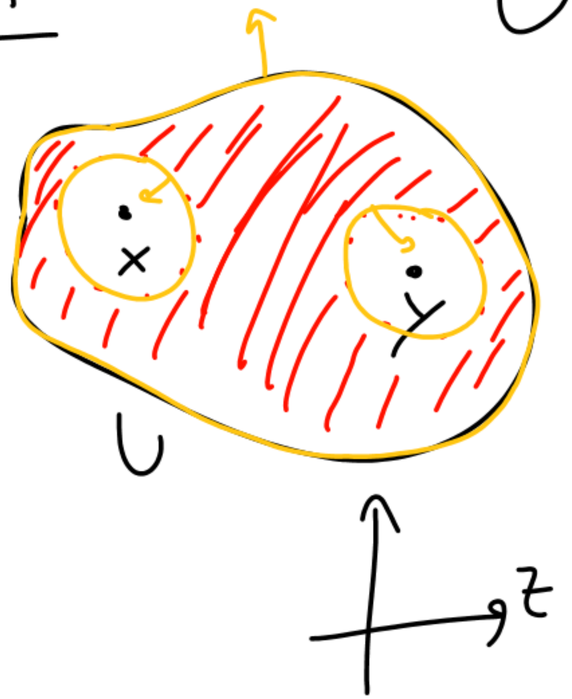
$$\Phi(|x|(y-\tilde{x})) = C_n\left(|x|\left|y - \frac{x}{|x|^2}\right|\right)^{2-n}$$

$$= C_n\left(\left||x|y - \frac{x}{|x|}\right|\right)^{2-n}$$

$$\lim_{x\to 0} \Phi(|x|(y-\tilde{x})) = \lim_{x\to 0} C_n\left(\left||x|y - \frac{x}{|x|}\right|\right)^{2-n} \longrightarrow C_n(1)^{2-n}$$

Thm $G(x,y) = G(y,x)$

Pf Green's identity with $G(x,z)$, $G(y,z)$



$$\iint_{U\setminus(B(x,\varepsilon)\cup B(y,\varepsilon))} \Big[G(x,z)\,\cancel{\Delta_z G(y,z)}^{\,0} - G(y,z)\,\cancel{\Delta_z G(x,z)}^{\,0} \Big]\, dz$$

$$= \iint_{\partial(U\setminus(B(x,\varepsilon)\cup B(y,\varepsilon)))} \Big[G(x,z)\,\frac{\partial G}{\partial \nu_z}(y,z) - G(y,z)\,\frac{\partial G}{\partial \nu_z}(x,z) \Big]\, dS_z$$

Note: $\partial\big(U\setminus(B(x,\varepsilon)\cup B(y,\varepsilon))\big) = \underline{\partial U} \cup \partial B(x,\varepsilon) \cup \partial B(y,\varepsilon)$

$$\int_{\partial U} \left[G(x,z) \frac{\partial G}{\partial \nu_z}(y,z) - G(y,z) \frac{\partial G}{\partial \nu_z}(x,z) \right] dS_z = 0$$

since $G(\cdot, z) = 0$ for $z \in \partial U$.

$$\int_{\partial B(x,\varepsilon)} \left[G(x,z) \frac{\partial G}{\partial \nu_z}(y,z) - G(y,z) \frac{\partial G}{\partial \nu_z}(x,z) \right] dS_z$$

As $\varepsilon \to 0$, the $\phi^x(z)$ & $\phi^y(z)$ parts are smooth, so they approach zero.

$$\int_{\partial B(x,\varepsilon)} G(x,z) \frac{\partial G}{\partial \nu_z}(y,z)\, dS_z = \int_{\partial B(x,\varepsilon)} \left[\Phi(x-z) \frac{\partial \Phi}{\partial \nu_z}(y,z) + \cancel{\phi^x(z) \frac{\partial \phi^y(z)}{\partial \nu_z}} - \Phi(x-z) \frac{\partial \phi^y(z)}{\partial \nu_z} - \cancel{\phi^x(z) \frac{\partial \Phi(y-z)}{\partial \nu_z}} \right]$$

$\to 0$ as $\varepsilon \to 0$

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$$\lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} G(x,z) \frac{\partial G}{\partial \nu_z}(y,z)\, dS_z = \lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} \Phi(x-z) \frac{\partial G}{\partial \nu_z}(y,z)\, dS_z$$

$$= \lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} c_n \varepsilon^{2-n} \frac{\partial G}{\partial \nu_z}(y,z)\, dS_z$$

$$\leq \lim_{\varepsilon \to 0} c_n \varepsilon^{2-n} \max_{B(x,\varepsilon)} |DG|\, \varepsilon^{n-1}$$

$$= 0.$$

Next term:

$$\lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} G(y,z) \frac{\partial G}{\partial \nu_z}(x,z)\, dS_z$$

$$= \lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} G(y,z) \frac{\partial \Phi(x-z)}{\partial \nu_z}\, dS_z - \overbrace{\int_{\partial B(x,\varepsilon)} G(y,z) \frac{\partial}{\partial \nu_z} \phi^x(z)\, dS_z}^{\to 0 \text{ as } \varepsilon \to 0}$$

$$= \lim_{\varepsilon \to 0} \int_{\partial B(x,\varepsilon)} G(y,z)\, C_n |x-z|^{1-n} (2-n)\, dS_z = \lim_{\varepsilon \to 0} \frac{C_n}{\varepsilon^{n-1}} \int_{\partial B(x,\varepsilon)} G(y,z)\, dS_z$$

$$= G(y,x)$$

Similarly, other term gives $-G(x,y)$

Add everything up: $G(y,x) - G(x,y) = 0$.

Energy Methods

Idea: Multiply your PDE by the solution (or some modification of it) and integrate over the domain.

Example Alternate proof of uniqueness for $\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U \end{cases}$

If u, v are C^1 solutions let $w = u - v$.

$$-\Delta w = -\Delta(u - v) = -\Delta u + \Delta v = f - f = 0$$

on ∂U, $w = u - v = g - g = 0$

$$-\Delta w = 0 \implies w\Delta w = 0$$

$$\int_U w\Delta w \, dx = 0$$

$$\int f f'' \, dx = -\int (f')^2 \, dx + \text{bdry} \ldots$$

$$\implies -\int_U |Dw|^2 \, dx + \int_{\partial U} w \frac{\partial w}{\partial \nu} \, dx = 0$$

In detail, $\operatorname{div}(wDw) = |Dw|^2 + w\Delta w$

Div Thm: $\int_U \operatorname{div}(wDw) \, dx = \int_U |Dw|^2 \, dx + \int_U w\Delta w \, dx$

$$\implies \boxed{\int_{\partial U} wDw \cdot \nu \, dx = \int_U |Dw|^2 + \int_U w\Delta w \, dx}$$

$$-\int_U |Dw|^2 dx + \cancel{\int_{\partial U} w \frac{\partial w}{\partial \nu} dS_x} = 0$$

$w = 0$ on ∂U

$$\int_U |Dw|^2 dx = 0 \Rightarrow |Dw|^2 = 0 \quad \text{a.e.}$$

If $u, v \in C^1(U) \Rightarrow w \in C^1(U)$

$\Rightarrow Dw = 0$ in U

$\Rightarrow w = \text{constant}$ in U

$w = 0$ on $\partial U \Rightarrow w = 0$ in U

$\Rightarrow u = v$.

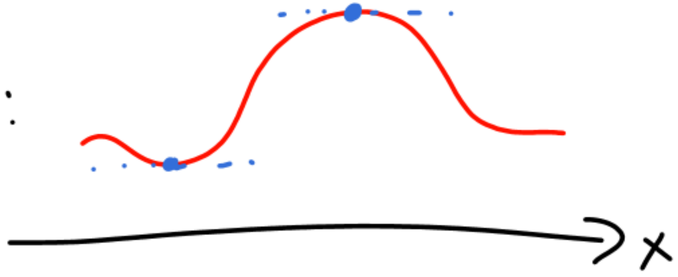
Thm If $u, v \in C^2(U) \cap C(\bar{U})$ both solve $-\Delta u = f$ in U, $u = g$ on ∂U, then $u = v$.

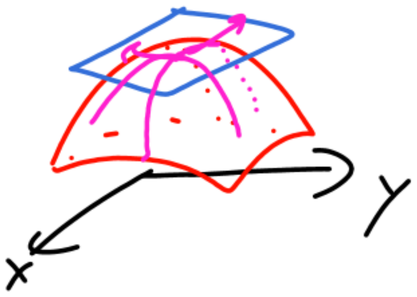
Dirichlet's Principle

Idea: Solutions of $-\Delta u = f$ are minimizers of an "energy functional" $I[w] = \frac{1}{2}\int_U [|Dw|^2 - wf]\,dx$

This is a problem in Calculus of Variations, the study of minimizing functionals.

Optimizing a function:





Extrema: $\frac{d}{ds} f(\vec{x}+s\vec{v})\Big|_{s=0} = 0$ for all $\vec{v}$

Optimize a functional $I[w]$

Extrema occur when $\frac{d}{ds} I[w+su]\Big|_{s=0} = 0$ for all admissible u.

<u>Thm</u> Let $I[w] = \int_U \left[\frac{1}{2} |Dw|^2 - wf \right] dx$

Let $\mathcal{A} = \{ w \in C^2(\bar{U}) : w = g \text{ on } \partial U \}$

u minimizes I over $\mathcal{A}$ if and only if

$-\Delta u = f$ in U and $u = g$ on ∂U.

<u>Pf</u> Assume u solves $-\Delta u = f$, $u = g$ on ∂U

Integrate PDE against $u - w$ for $w \in \mathcal{A}$.

$$-\int_U \Delta u\,(u-w)\,dx = \int_U f\,(u-w)\,dx$$

$$\text{Left side} = \int_U Du \cdot D(u-w)\,dx - \cancel{\int_{\partial U} \overbrace{(u-w)}^{=g-g=0} \frac{\partial u}{\partial \nu}\,dx}$$

$$\int_U Du \cdot D(u-w)\,dx = \int f(u-w)\,dx$$

$$\int_U Du \cdot D(u-w)\,dx = \int_U |Du|^2\,dx - \int_U Du \cdot Dw\,dx$$

Recall: $\vec{a}\cdot\vec{b} \leq |\vec{a}|\,|\vec{b}| \leq \frac{1}{2}\left(|\vec{a}|^2 + |\vec{b}|^2\right)$

$$\left(0 \leq (x-y)^2 = x^2+y^2-2xy \qquad xy \leq \tfrac{1}{2}(x^2+y^2) \quad \forall\, x,y \in \mathbb{R}\right)$$

$$\int_U |Du|^2\,dx - \int_U Du\cdot Dw\,dx = \int_U fu\,dx - \int_U fw\,dx$$

$$\int_U |Du|^2\,dx - \int_U fu\,dx = -\int_U fw\,dx + \int_U Du\cdot Dw\,dx \leq -\int_U fw\,dx + \int_U \frac{1}{2}\left(|Du|^2 + |Dw|^2\right)dx$$

Collect terms:

$$\int_U \left[\frac{1}{2}|Du|^2 - uf\right] dx \leq \int_U \left[\frac{1}{2}|Dw|^2 - wf\right] dx$$

i.e. $I[u] \leq I[w]$

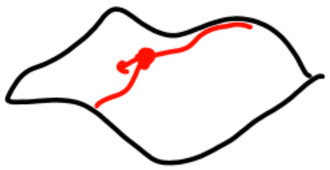
$w \in A$ was arbitrary $\Rightarrow$ u is a minimizer.

Conversely, assume u is a minimizer.

Then $I[u] \leq I[u+sv]$ for all $s \in \mathbb{R}$,

and all $v \in C^2(\bar{U})$ with $v = 0$ on ∂U.

($\Rightarrow$ $u + sv = g$ on ∂U)



It can be shown: $I[u+sv]$ is differentiable in s.

$$\Rightarrow \quad \frac{d}{ds} I[u+sv]\Big|_{s=0} = 0$$

$$\frac{d}{ds} I[u+sv] = \frac{d}{ds} \int_U \left[\frac{1}{2}|D(u+sv)|^2 - f(u+sv)\right] dx$$

$$= \int_U \left[(Du+sDv)\cdot Dv - fv\right] dx$$

$s=0$:

$$\int_U (Du\cdot Dv - fv)\,dx = 0$$

$$\int_U (Du \cdot Dv - vf)\,dx = 0$$

$v = 0$ on ∂U

$$\int_U (-\Delta u\, v - vf)\,dx + \cancel{\int_{\partial U} v \frac{\partial u}{\partial \nu}\,dS_x} = 0$$

$$\rightarrow \quad 0 = \int_U v(-\Delta u - f)\,dx \quad \text{for any } v \in C^2(\bar{U})$$

with $v = 0$ on ∂U.

This implies $-\Delta u - f = 0$ or $-\Delta u = f$.